BEING REDUCED IS LOCAL IN THE SMOOTH TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing phrase 'and we win' is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The general criterion for a property of schemes to be local in a topology. Same chapter, Lemma lemma-descending-properties, invoked at proof lines 5175 and 5186-5187. Reduces the claim to Zariski locality, preservation and descent.
- Reducedness is Zariski local. Schemes, Definition schemes-definition-reduced, invoked at proof lines 5176-5178. Gives the first clause.
- Smooth morphisms of affines preserve and reflect reducedness. Algebra, Lemmas algebra-lemma-reduced-goes-up and algebra-lemma-descent-reduced, invoked at proof lines 5179-5185. Gives the preservation and descent clauses.

Source reference. Original lemma 033F: sources/stacks-chow/tags.txt, source lines 4054–4054. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 033B: sources/stacks-chow/tags.txt, source lines 4050–4050. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0349: sources/stacks-descent/pinned-original-source.tex, source lines 4918–4918. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01J0: sources/stacks-chow/tags.txt, source lines 1997–1997. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(S) =$"S is reduced" is local in the smooth topology.*

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Proof. We will use Lemma 0349. First we note that "being reduced" is local in the Zariski topology. This is clear from the definition, see Schemes, Definition 01J0. Next, we show that if $S' \to S$ is a smooth morphism of affines and S is reduced, then S' is reduced. This is Algebra, Lemma 033B. Finally, we show that if $S' \to S$ is a surjective smooth morphism of affines and S' is reduced, then S is reduced. This is Algebra, Lemma 033F. Thus (1), (2) and (3) of Lemma 0349 hold and we win. □